Class 18

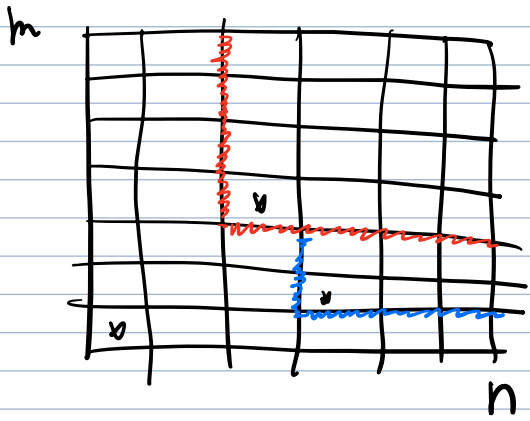
Theorem For every two player extensive form game (perfect information) with three outcomes player 1 wins, player 2 wins, or the game ends in a draw, only one of the following is true.

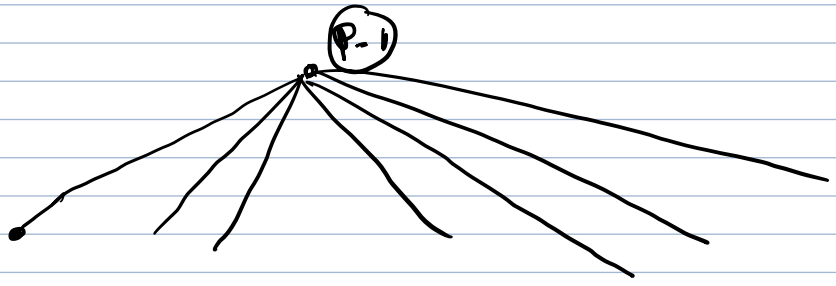
(i) P-1 has a winning strategy

(ii) P-2 has a winning strategy

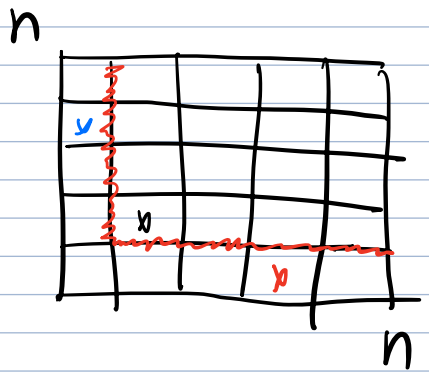
(iii) Both players have strategies guaranteeing at least a draw.

David Gale Chomp





$n = m$

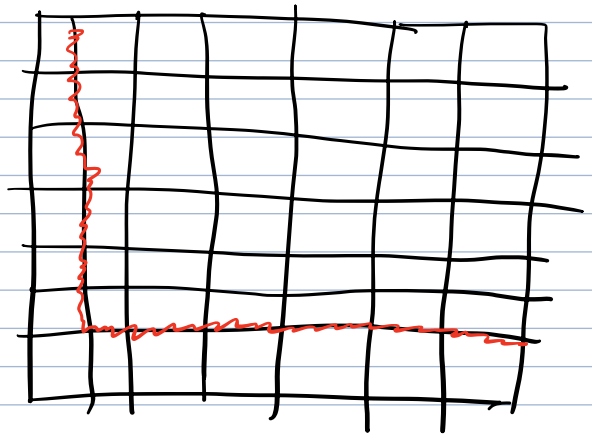


$n > 1$

Theorem For $n = m$, $P-1$ has a winning strategy. The strategy is as follows

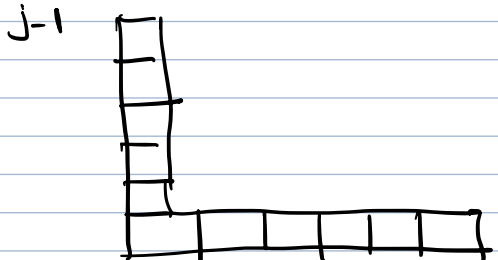
- first choose $(2, 2)$.
- Then do the opposite of $P-2$'s choice.

If $P-2$ chooses (i, j) th cell, choose the (j, i) th cell.



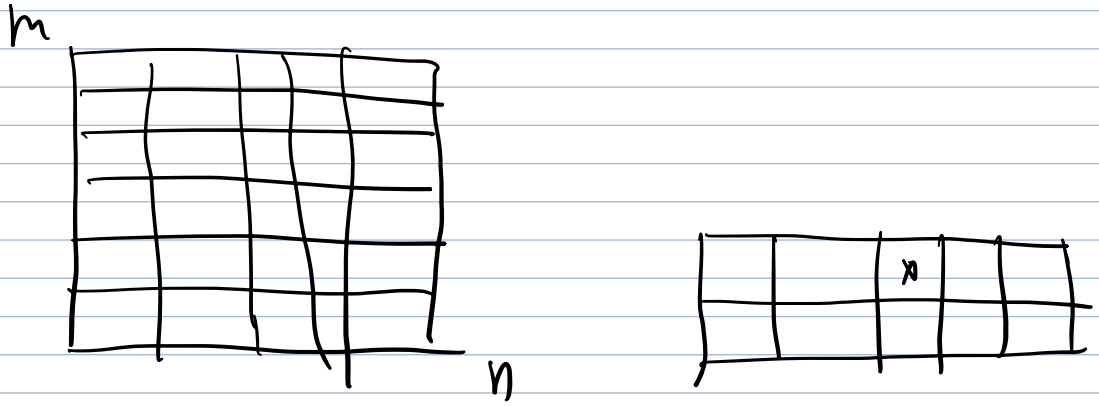
$\alpha \times \alpha$

$P_2 \to (i, j)$, $P_1 \to (j, i)$ th cell



i-1

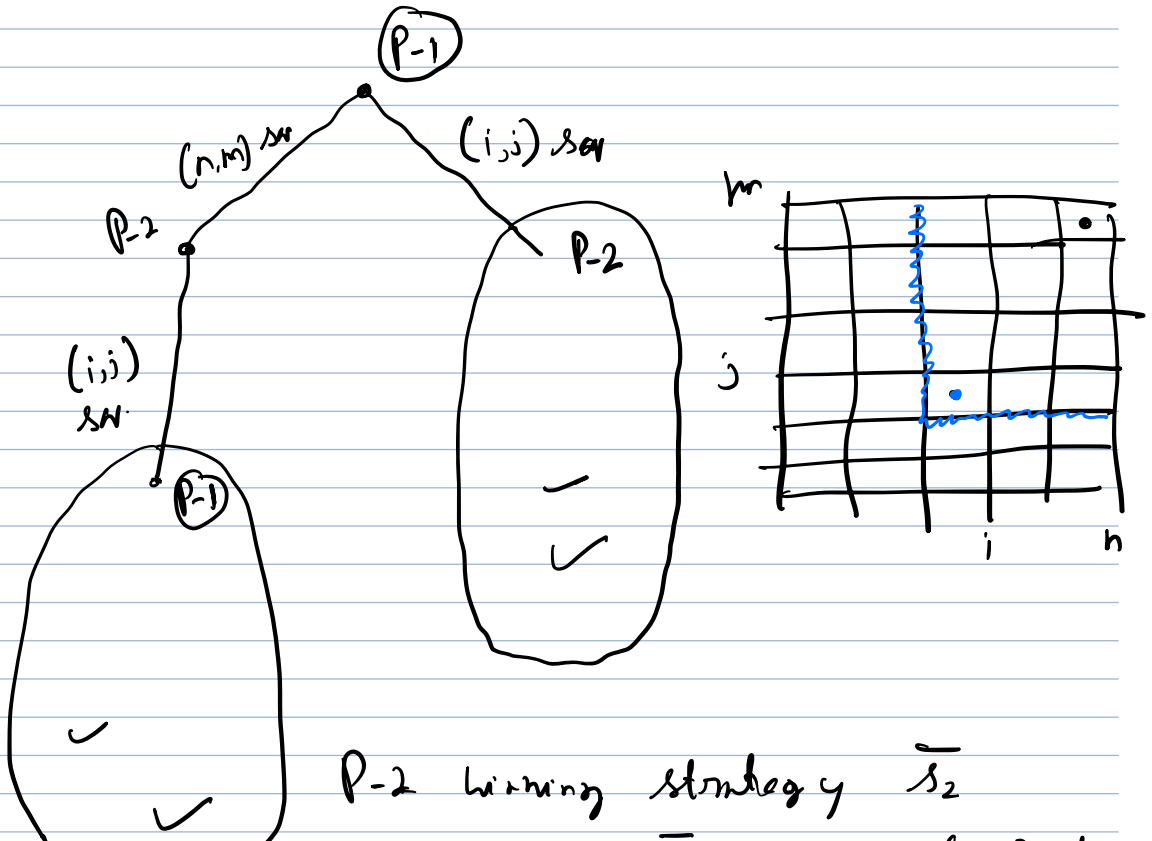
$n \neq n$



Theorem For $n \times m$ (where $n \neq m$), P-1 has always has a winning strategy.

Claim If P-2 has a winning strategy then P-1 has also a winning strategy.

Proof



P-2 winning strategy $\overline{S_2}$

and t_2 chooses (i,j) th square after P-1

P-1 winning strategy $\bar{s}_1$ chooses (n,m) th sq.

- first choose (i,j) th square and the follow $\bar{s}_2$ in that subgame starting from that position. And, do anything in all the other nodes.

This strategy $\bar{s}_1$ is a winning strategy for P-1.

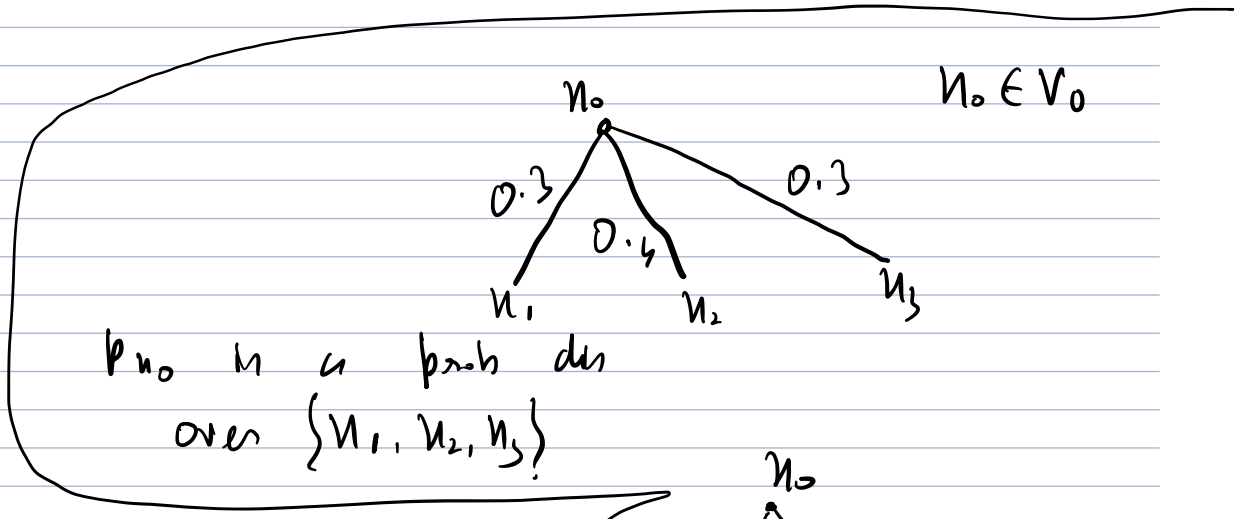
- $2 \times \infty$ board.
- $n \times \infty$ board $(n \geq 3)$

} You now cannot use the theorem.

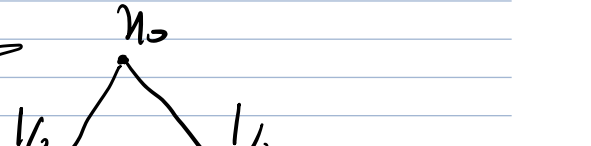
Check if any player has a winning strategy and find them

Defn A game in extensive form (with perfect information and chance moves) is a tuple

$$\Gamma = \langle N, V, E, n_0, (V_i)_{i \in N \cup \{0\}}, (p_n)_{n \in V_0}, (u_i)_{i \in N} \rangle$$



- $(V_i)_{i \in N \cup \{0\}}$ is a partition



$(\cdot)_{i \in N \setminus \{0\}}$

of the nodes that are not the leaves among the agents and the nature (P-0)



- $(p_n)_{n \in V_0}$ p_n is prob distribution over the the edges outgoing from n.